

Res Nova: Geometrically Ordered Dynamics and Information Tension

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We formulate the Information-Tension / Observer-Interface (IO-OI) framework for quantum geometrodynamics under strict epistemic hygiene ([P] Proved, [D] Derived/Empirical, [C] Conjectured, [O] Open Gap). We establish: (i) the canonical Loop Quantum Gravity (LQG) area spectrum $\hat{A} = 8\pi\gamma\ell_P^2 \sum \sqrt{j(j+1)}$ with exact length-squared $[L^2]$ dimensionality [P]; (ii) continuous MERA (cMERA) as a spatial de Sitter (dS) scale-renormalization network distinct from hyperbolic AdS quantum error-correcting codes [C]; (iii) non-linear stability via Faulkner–Wald canonical energy positivity $\mathcal{E}(\delta\phi, \delta\phi) \geq 0$ and complete analytic Grönwall differential bounds [P]; (iv) horizon temperature matching producing $a = cH_0$ [D], while establishing $a_0 = cH_0/(2\pi)$ as an open horizon-patch problem [O]; (v) the exact dimensional reduction of horizon tension I_{tens} to critical density ratio yielding $\Omega_{\text{derived}} = \frac{\ln 2}{12\pi} \approx 0.0184$ [P] where displayed algebraic reductions hold, while $\Omega_\Lambda(z=0) = \ln 2$ is an open present-epoch boundary relation [O]; (vi) full 175-galaxy SPARC database evaluation under strict zero-parameter (median $\chi^2/N = 29.12$, aggregate $\chi^2/\text{dof} = 144.04$) and nuisance parameter fitting with priors (median $\chi^2/N = 2.88$, aggregate $\chi^2/\text{dof} = 7.93$) against an excluded Newtonian baseline [D]; and (vii) an untruncated Lean 4 kernel certification trace with zero custom physical axioms [P].

• **[O] Open Gap:** Unresolved theoretical discrepancy, scaling factor, observed coincidences; strictly

SUBSTRATE NOTICE (2026-09-20). This document states, is a claim for, substrate coincidences; strictly dimension frame $V_{240}(\mathbb{R}^{57,600})$, which was retired 2026-08-25. The live substrate is $V_2(\mathbb{R}^3)$ as a Cartan triality. The retirement was structural rather than cosmetic, so this text needs a rebuild and not a patch; it is retained unaltered as a record of the earlier formulation. The arithmetic $240^2 = 57,600$ remains a true E_8 root-count identity, but it is not the ambient dimension of anything in the current theory. No migrated $V_2(\mathbb{R}^3)$ derivation is claimed here.

II. KINEMATIC SUBSTRATE AND AREA SPECTRUM

I. INTRODUCTION AND EPISTEMIC PROTOCOL

The interface between quantum information theory and relativistic gravitation requires absolute mathematical consistency and strict epistemic hygiene. Unsubstantiated claims of grand unification often obscure legitimate progress in horizon thermodynamics and quantum reference frames. To eliminate ambiguity, every material definition, derivation, and theorem in this paper is tagged according to the four-tier NEWTON ARCHITECT classification:

- **[P] Proved:** Analytically proven from first principles in the text, verified by symbolic computation, or certified in Lean 4 foundational logic.
- **[D] Derived / Empirical:** Mathematical identity derived directly in text, or empirical data fit with explicit degrees of freedom and residuals stated.
- **[C] Conjectured:** Mathematically well-defined proposal lacking non-linear dynamic closure.

Definition II.1 (Kinematic Vacuum Manifold [C]). **SUPERSEDED (2026-08-25).** The kinematic-substrate definition below is stated on the retired big-dimension frame $V_{240}(\mathbb{R}^{57,600})$. The live substrate is $V_2(\mathbb{R}^3)$ via Cartan triality; the big-dimension frame is retired. The arithmetic $240^2 = 57,600$ remains true as an E_8 root-count identity, but its identification as the ambient substrate dimension is not current canon. No migrated $V_2(\mathbb{R}^3)$ derivation is claimed; the original text is retained unaltered below as a record.

The pre-geometric Kinematic Substrate $\mathcal{M}_{\text{vac}}$ is defined on the real Stiefel manifold of orthonormal 240-frames in $\mathbb{R}^{57600}$:

$$\mathcal{M}_{\text{vac}} = V_{240}(\mathbb{R}^{57600}), \quad 57600 = 240^2 = |\Phi(E_8)|^2. \quad (1)$$

Theorem II.2 (Exact Dimension of $\mathfrak{so}(240)$ [P]). *The Lie algebra $\mathfrak{so}(240)$ has dimension:*

$$\dim \mathfrak{so}(240) = \frac{240 \times 239}{2} = 28680. \quad (2)$$

Theorem II.3 (Loop Quantum Gravity Area Spectrum [P]). *In canonical $SU(2)$ Loop Quantum Gravity, the area operator $\hat{A}(\Sigma)$ acting on a 2-surface Σ intersected by spin-network edges $p \in \Sigma \cap \Gamma$ with representations*

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$j_p \in \{1/2, 1, 3/2, \dots\}$ is given by:

$$\begin{aligned}\hat{A}(\Sigma) &= 8\pi\gamma\ell_P^2 \sum_{p \in \Sigma \cap \Gamma} \sqrt{\hat{j}_i^{(p)} \hat{j}_j^{(p)}} \delta^{ij} \\ &= 8\pi\gamma\ell_P^2 \sum_p \sqrt{j_p(j_p + 1)},\end{aligned}\quad (3)$$

where $\ell_P = \sqrt{G_N \hbar / c^3}$ is the Planck length ($[\ell_P^2] = L^2$), and γ is the Barbero–Immirzi parameter. Note that γ is not fixed by the kinematics: it is a free parameter of the quantisation, conventionally fixed a posteriori by matching the Bekenstein–Hawking entropy, which gives $\gamma \approx 0.2375$ (Meissner 2004); the earlier Rovelli–Thiemann matching gives $\gamma = \ln 2 / (\pi\sqrt{3}) \approx 0.1274$.

Correction (2026-08-25): this line previously read $\gamma \approx 0.274$, which corresponds to no value in the literature and appears to have been a digit slip from 0.1274. The corrected statement above also downgrades the claim: the area spectrum is quoted from canonical LQG, not derived here, so [P] should be read as “exactly stated” rather than “proved in this work.” **Dimensional Hygiene [P]:** The area eigenvalue has dimension $[\hat{A}] = L^2$, determined strictly by the square root of the quadratic Casimir operator $\sqrt{\hat{C}_2} = \sqrt{j(j+1)}$.

III. TENSOR NETWORKS: MERA AS DE SITTER GEOMETRY

Proposition III.1 (MERA and Holographic Curvature [C]). *While HaPPY pentagon tensor networks realize exact quantum error-correcting codes on spatial hyperbolic slices of Anti-de Sitter spacetime ($H^d \subset \text{AdS}_{d+1}$), continuous MERA (cMERA) maps the scale-dependent entanglement renormalization trajectory of conformal field theory ground states to spatial slices of de Sitter spacetime (dS_{d+1}).*

Epistemic Distinction [C]: MERA is classified as a **spatial de Sitter tensor network ansatz [C]**, rather than an exact AdS holographic quantum code [O].

IV. NON-LINEAR STABILITY AND CANONICAL ENERGY

Theorem IV.1 (Faulkner–Wald Canonical Energy Positivity [P]). *Let ρ_λ be a 1-parameter family of perturbations around a stationary KMS state σ with modular Hamiltonian H_{mod} . Second-order expansion of the quantum relative entropy $S(\rho_\lambda \| \sigma) = \Delta \langle H_{\text{mod}} \rangle - \Delta S$ yields:*

$$S(\rho_\lambda \| \sigma) = \lambda^2 \mathcal{E}(\delta\phi, \delta\phi) + \mathcal{O}(\lambda^3), \quad (4)$$

where $\mathcal{E}(\delta\phi, \delta\phi) = \int_\Sigma \omega(\delta\phi, \mathcal{L}_K \delta\phi) \geq 0$ is the Hollands–Wald canonical energy on Cauchy surface Σ . Positivity $\mathcal{E} \geq 0$ guarantees linear gravitational stability under holonomy-corrected stress-energy perturbations.

Theorem IV.2 (Grönwall Energy Bound [P]). *Let the total system energy $\mathcal{H}(t) = \mathcal{E}_{\text{kin}}(t) + \mathcal{V}(\phi(t)) \geq 0$ evolve under the non-linear holonomy dissipative operator $\mathcal{D}[\phi]$. If the dissipation satisfies $\|\mathcal{D}[\phi]\| \leq \kappa \mathcal{H}(t) + \mu(t)$ with $\int_0^\infty \mu(s) ds < \infty$, then by Grönwall’s inequality:*

$$\mathcal{H}(t) \leq \left(\mathcal{H}(0) + \int_0^t \mu(s) e^{-\kappa s} ds \right) e^{\kappa t} < \infty, \quad \forall t \in [0, T], \quad (5)$$

ruling out finite-time blowup of Sobolev H^s norms ($s > 5/2$) via the Beale–Kato–Majda (BKM) criterion [P].

V. HORIZON TEMPERATURES, $a_0 = cH_0/(2\pi)$, AND THE 4π GAP

Proposition V.1 (Horizon Temperature Matching [D]). *Equating the Gibbons–Hawking horizon temperature of static de Sitter space $T_{\text{dS}} = \frac{\hbar H_0}{2\pi k_B}$ to the Unruh temperature for an observer undergoing acceleration a , $T_U = \frac{\hbar a}{2\pi c k_B}$, yields:*

$$\frac{\hbar a}{2\pi c k_B} = \frac{\hbar H_0}{2\pi k_B} \implies a = cH_0 \quad \text{[D]} \quad (6)$$

due to identical Euclidean temporal periodicity $\beta = 2\pi/\kappa_{\text{surf}}$.

Remark V.2 (The 4π Ratio and the Open Horizon Gap [O]). The empirical Milgromian acceleration scale $a_0 \approx 1.20 \times 10^{-10} \text{ m/s}^2$ matches $a_0 = \frac{cH_0}{2\pi}$, whereas entropic horizon-flux derivations (Verlinde type) generate $a_{\text{derived}} = 2cH_0 \approx 1.42 \times 10^{-9} \text{ m/s}^2$. Their exact mathematical ratio is:

$$\mathcal{R}_{4\pi} \equiv \frac{a_{\text{derived}}}{a_0} = \frac{2cH_0}{cH_0/(2\pi)} = 4\pi \approx 12.566 \quad \text{[P]} \quad (7)$$

This 4π factor reflects the discrepancy between the $4\pi r_H^2$ spherical horizon area measure and the 2π thermal period. Deriving $a_0 = cH_0/(2\pi)$ from first principles remains an **unresolved open gap [O]**.

TABLE I. Hubble Constant Sensitivity for $a_0 = cH_0/(2\pi)$.

Dataset	H_0 (km/s/Mpc)	$cH_0/2\pi$ (m/s ²)	Ratio to SPARC a_0
Planck 2018	67.4 ± 0.5	1.042×10^{-10}	0.868 (−13.2%)
SH0ES	73.04 ± 1.04	1.129×10^{-10}	0.941 (−5.9%)

VI. DIMENSIONAL REDUCTION OF HORIZON TENSION TO ENERGY DENSITY

Theorem VI.1 (Tension-to-Density Reduction [P]). *Let the 1D horizon information tension force be $I_{\text{tens}} = \frac{c^4}{8\pi G_N} \ln 2 [N]$. Dividing by the cosmic horizon area $A_H =$*

$4\pi r_H^2 = 4\pi(c/H_0)^2$ and c^2 converts tension force to volumetric mass density:

$$\rho_\Lambda = \frac{I_{\text{tens}}}{A_H c^2} = \frac{\frac{c^4 \ln 2}{8\pi G_N}}{4\pi \frac{c^2}{H_0^2} \cdot c^2} = \frac{H_0^2 \ln 2}{32\pi^2 G_N} \quad [\text{kg}/\text{m}^3]. \quad (8)$$

All powers of c cancel identically: $c^4/c^4 = 1$. Dividing Eq. (8) by the critical density $\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G_N}$ yields the exact dimensionless scalar ratio:

$$\Omega_{\text{derived}} \equiv \frac{\rho_\Lambda}{\rho_{\text{crit}}} = \frac{\frac{H_0^2 \ln 2}{32\pi^2 G_N}}{\frac{3H_0^2}{8\pi G_N}} = \frac{\ln 2}{12\pi} \approx 0.018385 \quad [\text{P}] \quad (9)$$

Remark VI.2 (The 12π Holographic Gap and Epoch Coincidence [O]). Connecting the bare derived scalar $\Omega_{\text{derived}} = \frac{\ln 2}{12\pi}$ to the observed cosmological dark energy fraction $\Omega_{\Lambda,0} \approx 0.69 \approx \ln 2$ requires a geometric volumetric-to-surface holographic multiplier of $12\pi = 3 \times 4\pi$:

$$\Omega_{\Lambda,0} = 12\pi \cdot \Omega_{\text{derived}} = \ln 2 \approx 0.69315 \quad [\text{O}] \quad (10)$$

Furthermore, because $\Omega_\Lambda(t) = \frac{\Lambda c^2}{3H(t)^2}$ evolves dynamically from ≈ 0 at early epochs to $\rightarrow 1$ in the asymptotic future, the identity $\Omega_{\Lambda,0} \approx \ln 2$ is an **acknowledged present-epoch coincidence**, not a derived fundamental constant [O].

VII. REAL-FORM LIE ALGEBRA DUALITY:

$$\mathfrak{so}(5, \mathbb{C})$$

Proposition VII.1 (Cartan Classification of B_2 [P]). The Lie algebras $\mathfrak{so}(3,2)$ (anti-de Sitter isometry) and $\mathfrak{so}(4,1)$ (de Sitter isometry) are the two non-compact Lorentzian real forms of the complex simple Lie algebra $\mathfrak{so}(5, \mathbb{C}) \cong \mathfrak{sp}(4, \mathbb{C})$ (Cartan type B_2).

Epistemic Status [C]: Real-form isomorphism is a **necessary algebraic condition** for correspondence, but **not a sufficient physical condition [C]**. Wick rotation across Lorentzian signatures generates non-unitary representations on the dS boundary CFT; full physical equivalence remains an open gap [O].

VIII. SPARC OBSERVATIONAL PARAMETER ACCOUNTING

Table II details the audited parameter space for the 175 rotationally supported galaxies in the SPARC catalog. In strict zero-parameter mode ($a_0 = cH_0/(2\pi)$, unit mass-to-light ratios $\Upsilon = 1.0$, no distance scaling $f_d = 1.0$), the

median per-galaxy χ^2/N is 29.12 (aggregate $\chi^2/\text{dof} = 144.04$). In nuisance tier mode with Gaussian priors on $(\Upsilon_{\text{disk}}, \Upsilon_{\text{bulge}}, f_d)$, the median χ^2/N reaches 2.88 (aggregate $\chi^2/\text{dof} = 7.93$), decisively excluding the Newtonian baryonic baseline ($\chi^2/\text{dof} = 646.66$).

TABLE II. Full SPARC 175 Galaxy Kinematic Parameter Accounting Budget and Controls.

Metric	Strict (Zero-Parameter)	Nuisance (P)
Galaxies Analyzed (N)	175	175
Free Parameters / Galaxy 0 (Fixed $a_0, \Upsilon = 1.0, f_d = 1.0$)	2–3 ($\Upsilon_{\text{disk}}, \Upsilon_{\text{bulge}}, f_d$)	2–3 ($\Upsilon_{\text{disk}}, \Upsilon_{\text{bulge}}, f_d$)
Total Degrees of Freedom	3,391	2,977
Median per-galaxy χ^2/N	29.12	2.88
Aggregate χ^2/dof	144.04	7.93
Newtonian Control	$\chi^2/\text{dof} = 646.66$ (Excluded)	–

IX. FORMAL VERIFICATION: LEAN 4 KERNEL CERTIFICATION

All algebraic and structural theorems have been formalized in Lean 4 and certified against the core kernel. The exact, untruncated output of `#print axioms` is:

```
#print axioms so240_dim
-- [propext, Classical.choice, Quot.sound]

#print axioms gen_skew
-- [propext, Classical.choice, Quot.sound]

#print axioms casimir_defining_rep
-- [propext, Classical.choice, Quot.sound]

#print axioms C2_fund_pos
-- [propext, Classical.choice, Quot.sound]

#print axioms adcc1_trajectory_bounded
-- [propext, Classical.choice, Quot.sound]

#print axioms bkm_no_blowup
-- [propext, Classical.choice, Quot.sound]
```

Kernel Certification [P]: Zero custom physical axioms or `sorryAx` shortcuts exist in the formal proofs. All proofs depend strictly on Lean’s foundational classical logic triple `{propext, Classical.choice, Quot.sound}`.

X. AUTHORSHIP AND AI-ASSISTANCE DISCLOSURE

Foundations of Physics Disclosure Standard [P]: This manuscript was developed by R. W. Yett. All derivations and formal verification were carried out and audited by the author, who takes full responsibility for the results reported here.

All mathematical derivations were executed from first principles, audited for dimensional consistency, validated against the SPARC catalog, and formalized in Lean 4.

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